

CSIS 4495 – Applied Research Project

Progress Report 2

GREENPATH – Sustainable Products Website with AI Chatbot Integration

FRANCIS AWOYALE (300356743)

Work Date/Hours Log

The table below show summary for the work done till date with the respective time spent on each task.

Date	Number of Hours	Description of Work Done
2025-09-25	3	Tested AWS Lambda and DB connectivity using Secrets Manager and also verified the JSON responses.
2025-09-26	2	Fixed API 500 errors by correcting VITE_API_URL then added CORS headers in Lambda.
2025-09-27	1	Cleaning of the Emission dataset and combining the sector cumulative data.
2025-09-28	3	Integrated API data into Analytics page and validated the chart outputs.
2025-09-29	2	Resolved Amplify deployment errors (React Router 404 & MIME type issues).
2025-10-30	2.5	Re-structured the Analytics.tsx for Line and Bar visualization switching and error handling.
2025-10-01	3	Added RootLayout and global green and white theme. Also, improved navigation size and contrast.
2025-10-02	3	Designed Home page carousel and added introductory text on sustainability to enhance UI.
2025-10-03	3	Enhanced Analytics UI with dataset summary and CO ₂ impact section.
2025-10-04	3	Built Categories dashboard with icons and clickable cards.
2025-10-05	3	Created Category Detail page with Hero and gallery lightbox.
2025-10-06	2	Fixed blurry images by adding responsive src set. Also optimized image loading.
2025-10-07	2	Improve UI with Standardized fonts and spacing across all pages.
2025-10-08	2	Compiled Progress Report 2 and updated work log.

Description of Work Done

These past weeks, I worked on completing the full integration between the backend API and the Analytics page, refining the frontend design, and optimizing the dataset for performance. I configured AWS Lambda to securely connect to the PostgreSQL database through AWS Secrets Manager, corrected the `VITE_API_URL` variable, and added proper Cross-Origin Resource Sharing (CORS) headers to fix repeated 500 and 404 API errors. I also cleaned and combined the emissions dataset by aggregating cumulative sector data by year and country, ensuring it matched the Analytics structure and loaded faster without slowing the page. Deployment issues on AWS Amplify, such as blank pages and Multipurpose Internet Mail Extensions (MIME) type errors, were resolved by enabling single page application rewrites and clearing old build caches. On the frontend, I added a `RootLayout` with a consistent green and white theme, redesigned the Home page with a sustainability carousel, and expanded the Analytics page to include a dataset description above the charts. I also built the `Categories` and `CategoryDetail` pages with hero images and gallery lightboxes, fixing earlier blurriness using responsive image sets. Currently working on adding images to the `CategoryDetail` pages. This updated work done so far has improved both performance and usability, resulting in a stable, well-styled, and fully functional GreenPath web application.

Repo Check in of Implementation Completed

The files/folders I have checked in the repo since report 1 are as follows:

Recent commits under `Implementation/greenpath/` include:

`src/layout/RootLayout.tsx` – New global layout and theme for consistent navigation.

`src/pages/Home.tsx` – Redesigned with sustainability carousel and introductory text.

`src/pages/Analytics.tsx` – Connected live API data and added dataset summary above charts.

`src/pages/Categories.tsx` / `CategoryDetail.tsx` – Clickable category cards with hero and gallery lightbox.

`backend/lambda/index.js` – AWS Lambda function handling `/aggregate` API requests, database connection, and CORS headers.

`backend/lambda/package.json` – Supporting dependencies and environment setup for the Lambda function.

`public/images/categories/` – Re-organized image assets for hero and gallery sections.